

PRAVALIKA THOTA

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SUMMARY

Data Analyst experienced in preparing, validating, reconciling, and analyzing enterprise datasets using Excel, SQL, Python, Power BI, and Tableau. Skilled in data cleaning, reporting, data quality improvement, and developing accurate datasets for operational reporting and enterprise system implementations.

EDUCATION

MS in Business Analytics, California State University, East Bay, CA, USA	08/2023 – 05/2025
MA in Interior Architecture and Design, Heriot-Watt University, Dubai, UAE	08/2019 – 10/2021
Bachelor of Technology in Civil Engineering, VJCET, India	07/2013 – 05/2017

TECHNICAL SKILLS

Programming	Python, SQL
Excel	Excel (Advanced Formulas, XLOOKUP, VLOOKUP, INDEX-MATCH, Pivot Tables, Conditional Formatting, Data Validation, CSV Processing)
Visualization	Tableau, Microsoft PowerBI
Databases	MySQL, Snowflake
Data Analysis	Data Cleaning, Data Validation, Data Reconciliation, Data Profiling, KPI Reporting, Statistical Analysis
Data Engineering	ETL, Airflow
Cloud	AWS

WORK EXPERIENCE

Data Engineer (Reporting & Analytics), Delta Dental, Remote 10/2025 – Present

- Partnered with business stakeholders to translate reporting requirements into 5+ reusable analytical datasets, accelerating recurring KPI reporting and enabling faster operational decision-making.
- Prepared and validated enterprise datasets containing 3K+ records, improving reporting consistency and reducing downstream data quality issues by 20%.
- Identified reporting discrepancies through data profiling, reconciliation, and trend analysis, supporting performance improvement initiatives and increasing confidence in business metrics.
- Resolved inconsistencies across multiple source and target datasets by investigating missing, duplicate, and mismatched records, improving reporting accuracy and stakeholder trust.
- Developed SQL and Python workflows to cleanse, standardize, and transform enterprise datasets, producing analysis-ready data for reporting and operational analytics.
- Implemented automated validation checks for record counts, duplicates, null values, schema consistency, and business-rule compliance, reducing reporting errors by 20% and improving overall data quality.
- Optimized complex SQL queries through join optimization, indexing strategies, and early-stage filtering, decreasing report preparation time by 30% and improving analyst productivity.
- Collaborated with analytics, technical, and business teams to define reporting requirements and deliver reusable datasets, accelerating insight generation across multiple departments.
- Documented data transformation logic, validation procedures, and reporting workflows, reducing onboarding time by 25% while improving process consistency.

Graduate Teaching Assistant, California State University, East Bay, US 02/2025 – 05/2025

- Reviewed over 200 Python and SQL assignments, identifying logical and database design errors while maintaining consistent grading standards across multiple course sections.
- Evaluated 70+ advanced SQL assignments covering joins, CTEs, stored procedures, constraints, and subqueries, reinforcing database best practices and query optimization principles.
- Assessed database designs for 35 students, validating ERDs, normalization, transactions, and data integrity concepts to ensure adherence to relational database standards.
- Performed plagiarism detection across 200+ submissions, collaborating with faculty to maintain grading accuracy and academic integrity.

Data Analyst, Linkfields Innovation 05/2018 – 05/2019

- Analyzed 10K+ business records using SQL, Python, and Excel to uncover operational trends and performance gaps, supporting 3 strategic business initiatives and data-driven decision-making.
- Integrated and reconciled data from three enterprise systems, reducing manual preparation by 25% while improving

reporting consistency across recurring business reports.

- Developed advanced SQL queries using joins, CTEs, window functions, and aggregations to automate 5 recurring analytical requests, decreasing report turnaround time by 25%.
- Automated recurring reporting and data preparation workflows using SQL and Python, reducing manual effort by 25% while improving reporting accuracy and efficiency.
- Designed 3 interactive Power BI and Tableau dashboards tracking 8 business KPIs, improving executive visibility into operational performance and enabling faster business decisions.
- Validated enterprise datasets through reconciliation, duplicate detection, anomaly analysis, and business-rule validation, preventing inaccurate data from reaching downstream reports and dashboards.
- Consolidated historical data across three business systems, identifying operational trends and data inconsistencies that supported process improvement initiatives.
- Collaborated with cross-functional business and technical teams to translate reporting requirements into reusable SQL analyses, KPI definitions, and standardized reporting datasets.
- Standardized SQL queries and reporting templates across 5 recurring reports, reducing duplicate development effort by 20% while improving reporting accuracy and consistency.

PROJECTS

Insurance Claim Prediction

- Analyzed historical insurance claims data using Python and SQL to identify cost drivers, supporting predictive modeling and business decision-making.
- Engineered predictive features and optimized CatBoost models, achieving an MAE of 730 and improving prediction accuracy over baseline models.
- Evaluated 10+ regression algorithms, selecting the highest-performing model and improving prediction reliability through comparative performance analysis.
- Identified key claim cost drivers through statistical analysis and feature importance evaluation, providing actionable insights for insurance risk assessment.

Data Pipeline Automation with Apache Airflow and Bash

- Developed an automated ETL pipeline using Bash and Apache Airflow to consolidate structured data from multiple sources, eliminating manual processing.
- Designed an Airflow workflow to automate extraction, transformation, and loading tasks, improving data availability for downstream reporting.
- Standardized CSV, TSV, and fixed-width datasets through automated cleaning and transformation, producing consistent analysis-ready datasets.

Cloud Data Ingestion with AWS S3 and Snowflake

- Built a cloud-based ingestion pipeline using AWS S3 and Snowflake, automating enterprise data loading for reporting and analytics.
- Implemented incremental processing using Snowflake Streams and Tasks, ensuring continuously updated datasets while minimizing redundant processing.
- Developed optimized analytical tables for SQL reporting, improving dashboard performance and supporting faster business analysis.

Yoga pose prediction from videos

- Engineered pose-based features from MediaPipe keypoints to create structured datasets for predictive analysis.
- Evaluated Conv1D and LSTM models across five yoga poses, achieving 87% classification accuracy through feature engineering and model optimization.

Airbnb Price Optimization

- Analyzed Airbnb listing data using Python to identify pricing trends across 5000 properties, supporting data-driven pricing recommendations.
- Cleaned and transformed property, booking, and location datasets, improving data quality for exploratory analysis and predictive modeling.
- Developed predictive pricing models achieving 87% accuracy, enabling more accurate estimation of optimal listing prices.
- Presented analytical findings through visualizations and statistical summaries, translating complex data into actionable business recommendations.

CERTIFICATIONS

[Python for Everybody Specialization](#),

[Tableau BI Analyst](#), [Power BI](#), [Cognos Analytics and Looker Studio](#)

[Business Statistics and Analysis Specialization](#)

[BI Foundations with SQL, ETL and Data Warehousing](#)
[ETL and Data Pipelines with Shell, Airflow and Kafka](#)
[Explore Data Concepts in Microsoft Azure](#)